# EVRYN Overarching Three-Phase Research Scope & Monograph Demarcation

## PHASE 1

**Multi-Page Causal Extraction**

DELIVERED (Phase 1.14)

- Full-text PMC-OA JATS XML & PDF parsing with geometry
- Section discourse tagging & coordinate tracking
- Syntactic & agent-nominal causal trigger parsing
- Dual-sentence context resolution [Sc, Se]
- Bounded transitive multi-hop reasoning (<= 2 hops)
- Calibrated confidence verification & Platt scaling
- 100% Provenance contract (Doc, page, offsets, quotes)
- 20-Paper authentic PMC benchmark validation

Grounded Graphs →

## PHASE 2

**Corpus Transfer & Adaptation**

IN PROGRESS

- Multi-document graph fusion across literature
- Cross-paper entity normalization & alias resolution
- 25-Year longitudinal corpus trajectory tracking
- Causal mechanism drift & belief revision detection
- Continual SCM adaptation with zero forgetting
- Conformal semantic gating via PubMedBERT
- Empirical evaluation on temporal SCM streams
- Cross-study consensus & refutation modeling

Longitudinal Corpus →

## PHASE 3

**Prospective Discovery Prediction**

PLANNED

- Dynamic causal graph manifold representation
- Prospective emergent link forecasting
- Pre-publication hypothesis generation & ranking
- Multi-hop cross-domain mechanism completion
- Prospective validation on post-cutoff publications
- Biomedical discovery prioritization pipeline
- Comprehensive M.Tech defense monograph
- Reproducible open-source software release